
CF000: COMPLETE FORMALISM - PRIMITIVE DISTINGUISHABILITY AND ORTHOGONAL ARTICULATION

COMPLETE FORMALISM

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ABSTRACT

This paper develops a root formalism for how distinguishability can arise without presupposing geometry, locality, carrier structure, calculus, or later physical machinery. By excluding isolated nullity and isolated undifferentiated totality as sterile primitive terminals, it derives a unique admissible origin as unresolved primitive opposition. Because discharge of this opposition would collapse into already excluded terminal states, any admitted articulation must bear the invariant internally. This yields non-flat distinguishability as the first live consequence. The paper then shows that articulation within a single admitted axis is exhaustively saturable but not terminal: once no genuinely new invariant-bearing articulation remains available in that same axis, continuation there is closed while discharge remains impossible. Orthogonal re-articulation is therefore forced. In this way the paper closes the primitive route from 0D to 1D to 2D and identifies the two-axis stage as the first point at which algebra becomes admissible. The result is a self-contained root formalism that prepares, but does not yet assume, the carrier and induced geometric machinery developed in the next paper.

Keywords primitive origin · distinguishability · invariant-bearing articulation · orthogonal re-articulation · same-axis saturation · algebra unlock

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Governing Rule of This Document

This CF is the root written source of truth for the pre-carrier, pre-differential layer of the present VDM branch. The governing claim of this document is the single fundamental primitive bifurcation law: the only logically admissible primitive origin is one primitive condition whose content bears two mutually unreconcilable terminal poles, absolute nullity and absolute undifferentiated totality; neither isolated pole can terminate the burden without collapse; unresolved pole-opposition is therefore constitutive of the admissible origin; every admitted articulation stands under that same undischarged burden. This rule is not a loose metaphor, a one-time split, or a dynamical picture. It is the manuscript basis. The paper must therefore read as one invariant spine: forced admissible origin, first non-flat articulation, higher-resolution admission, recursive restatement, saturation of available lawful expression, and orthogonal re-articulation into 2D are not separate sources. They are more articulated determinations of one undischarged primitive invariant.

Consequence of this rule

For this document:

- the primitive opposition must be treated as derived from the only admissible origin, not inserted as a detachable extra assumption,
- the origin must be treated as one primitive condition, not as two externally separated things,
- the two poles must be treated as logically distinct but primitively unapart in being,
- non-bearing of the invariant at any admitted articulation is contradictory,
- apparent settlement short of terminal discharge is not final resolution,
- CF000 closes the post-origin route 0D -> 1D -> 2D together with the first axis-proper effective invariants at 1D and 2D: continuous same-axis bifurcation and the numerical relation-grid invariant.
- CF00 must therefore begin from the inherited post-2D invariant stack proved here and derive the first lawful host/carrier consequences of that closure.
- Carrier construction, QGT, and metriplectic naming still do not belong to CF000.

Relationship to Canon and External Documents

Canon documents, proposal artifacts, notes, and later CFs may be cited for provenance, naming, alignment, and broader program integration. They may not be used to outsource root derivation that belongs here. For the present manuscript:

- H000 preserves the proposal-grade primitive bifurcation hypothesis,
- the doctrinal note on the single fundamental primitive bifurcation law preserves the compact intended root law,
- A8 is not a primitive generator here; it is a later-regime quantitative expression of the same invariant under added carrier burden,
- CF00 is downstream of the closure obtained here,
- CF01 remains later still and is not licensed inside the theorem-bearing spine of CF000.

No later canon object - carrier-family, QGT, metric curvature, Laplacian, differentiability bridge, or named $J \oplus M$ split - may be used to prepay burden at the root layer.

Relationship to the CFN

No paired CFN is required for theorem closure at the present stage. If a later CFN is built, it may only instantiate, test, or illustrate constructions already formalized here. It may not supply missing root logic. A good rule for this stage is: The CF proves and defines. A future CFN may execute and illustrate.

Executive Summary

This Complete Formalism formalizes the CF000 root branch beneath CF00 at the level of the staged route 0D -> 1D -> 2D. It treats H000 as the proposal-grade antecedent and turns the admissible portion of that root architecture into

theorem-bearing closure at this stage. It classifies the only admissible primitive origin, excludes terminal discharge into absolute nullity and absolute undifferentiated totality, derives the first non-flat invariant-bearing articulation, closes higher-resolution admission and recursive restatement, derives the first pre-metric orthogonal rotation, and closes the second orthogonal rotation into 2D. What is primitive here is only the terminal poles, their opposition, realizability discipline, flatness discipline, and the one-origin invariant frame. Geometry, locality, carrier structure, QGT, metric structure, Laplacian structure, and differentiability are not primitive here and are not proven here. The decisive result of CF000 is that the invariant cannot be terminally satisfied within the currently admitted axis of articulation. Once lawful expression within that axis is exhausted, terminal discharge is still impossible, so an orthogonal re-articulation into a new domain is forced. The 1D->2D instance of that logic is the closure of this document. That second orthogonal rotation is the first point at which algebra is licensed.

At first admission, the invariant does not merely occupy an axis/domain as a passive container condition. The admitted 1D axis/domain immediately acquires a new effective invariant proper to that axis/domain: continuous same-axis bifurcative articulation under the inherited primitive invariant. Thus 1D already bears the invariant twice: first as inherited primitive unresolved burden, and second as the axis-proper continuity/bifurcation law of its own admitted expression. Seriality, repeatable more/less, and proto-natural-number structure therefore become licensed already at 1D. When 2D is admitted, the prior invariant stack is inherited rather than erased, and the newly admitted second axis/domain immediately acquires its own effective invariant as well. That 2D axis-proper effective invariant is named the numerical relation-grid invariant: inherited 1D bifurcation and new second-axis bifurcation jointly generate a two-axis articulation field in which numerical relations become lawfully expressible. That is the structural reason algebra first becomes admissible only at 2D.

Principal deliverables of this CF:

- primitive origin classification and forced admissible-origin theorem,
- exclusion of nullity, undifferentiated totality, and any third terminal primitive pole,
- derivation of differentiated articulation and the lowest-resolution invariant-bearing degree,
- closure of HRDCL, HRDAP, contextual noncoincidence, apartness, ORS, and RRP,
- closure of the first axis-proper effective invariant at 1D as continuous same-axis bifurcative articulation,
- seriality, repeatable more/less, and proto-natural-number structure as derived consequences already at 1D,
- closure of same-domain saturation and the forced second orthogonal rotation into 2D,
- closure of inherited invariant stacking at 2D together with a new second-axis effective invariant named the numerical relation-grid invariant,
- first numerical relation-grid structure at 2D,
- algebra unlock at the 2D stage,
- explicit stage-correct handoff statement to CF00,
- validation logic and non-load-bearing empirical consistency notes for the finite-domain/A8 regime.

Read Me First: Writing Rules for This CF

1. This document is self-contained at the level required to understand, test, and reuse the formalism declared here.
2. Every theorem-bearing section exposes its proof burden directly.
3. No section may hide essential logic behind phrases such as “by standard arguments,” “similarly,” or “left to later canon.”
4. All dependency order must be honest. No theorem may use objects that have not yet been defined or derived.
5. Distinguish clearly between:
 - primitive,
 - proved in this CF,
 - derived structural consequence,
 - empirical consistency note,
 - forward note for CF00,
 - interpretive extension.

6. This document uses pre-metric orthogonality only in the minimal sense proved here: irreducibility of a new axis/domain relative to the exhausted current one. No Euclidean metric is smuggled in.
7. If a section is incomplete, the CF is incomplete.

1. Scope, Ontology, and Primitive Commitments

1.1 Root question

CF00 begins later than the absolute root. CF000 exists because that is already too late for an origin account. The root question of this document is: What is the earliest admissible primitive law from which distinguishability, higher articulation, and the first lawful orthogonal re-articulation can be earned without primitive geometry, carrier structure, locality, time, causality, valuation, parameterization, or differentiability?

1.2 Declared closure and stage gate

This CF closes only the staged route:

0D -> 1D -> 2D. Here that means:

- 0D: primitive unresolved origin before admitted axis-specific articulation,
- 1D: first admitted axis of non-flat invariant-bearing articulation,
- 2D: the first orthogonal re-articulation forced when the 1D axis is exhausted without discharge. At each admitted axis/domain within this closure, the invariant is not merely preserved but re-expressed as a new axis-proper effective invariant proper to that admission: first at 1D as continuous same-axis bifurcation and again at 2D as the numerical relation-grid invariant. This CF does not prove QGT, metric geometry, carrier-family mechanics, differentiability, or the named conservative/dissipative split. Those belong later.

1.3 Primitive ontology

Primitive in this rewrite:

- two distinguished terminal pole marks,
- their non-identity,
- their sterility for realized differentiated articulation,
- their opposition,
- an origin-candidate predicate,
- a non-vacuousness predicate,
- a realizability predicate,
- a flatness predicate,
- and the one-origin invariant discipline. Derived in this rewrite:
- primitive invariant carriage,
- differentiated articulation,
- lowest-resolution invariant-bearing degree,
- genuine higher-resolution degree,
- recursive restatement,
- contextual noncoincidence,
- apartness,
- pre-metric orthogonal re-articulation,
- same-axis saturation,
- the 1D axis-proper effective invariant,
- continuous same-axis bifurcation,

- seriality / proto-natural-number structure at 1D,
- inherited invariant stacking at 2D,
- the 2D axis-proper effective invariant named the numerical relation-grid invariant,
- first numerical relation-grid structure,
- 2D closure,
- algebra unlock.

1.4 Forbidden primitive imports

The following are forbidden as primitive in CF000:

- carrier language,
- topology, metric structure, or chart language,
- locality or neighborhood structure,
- time parameter, event sequence, propagation, or causality,
- graph, node, edge, or tree language,
- QGT,
- named J , M , or $J \oplus M$ machinery,
- differentiability, Laplacian, or smooth support structure.

1.5 Status convention

Every major claim in this document is labeled as one of:

- Proved in this CF
- Derived structural consequence
- Empirical consistency note (non-load-bearing)
- Forward note for CF00
- Interpretive / out-of-scope extension

2. Mathematical Setting and Definitions

2.1 Logical domain of discourse

Let C denote the logical domain of candidate realization conditions. This is not a primitive ontology of objects in a world. It is only a logical range of discourse, introduced so that realizability claims can be stated without presupposing points, carriers, graphs, or manifolds.

2.2 Definitions

Absolute nullity. Complete realized absence. Formal symbol: distinguished pole mark 0. Status: Primitive. Absolute undifferentiated totality. Complete homogeneous one-ness without internal noncoincidence. Formal symbol: distinguished pole mark 1. Status: Primitive. Origin candidate. Candidate primitive condition for a realized branch. Formal symbol: predicate $\text{Orig}(x)$ on C . Status: Primitive. Sterile. Cannot admit realized differentiated articulation while remaining what it is. Formal symbol: predicate $\text{Ster}(x)$. Status: Primitive. Flat. Contains no realized internal noncoincidence by which differentiated articulation could be sustained. Formal symbol: predicate $\text{Flat}(x)$. Status: Primitive. Non-vacuous. Not complete realized absence in the root sense. Formal symbol: predicate $\text{NV}(x)$. Status: Primitive. Realizable. Admissible as a non-vacuous realized branch-condition. Formal symbol: predicate $\text{Real}(x)$. Status: Primitive.

Opposition. Non-identity of the two terminal poles together with their shared sterility verdict. Formal symbol: predicate $\text{Opp}(x, y)$. Status: Primitive. Primitive invariant. The one admissible primitive origin-condition bears unresolved pole-opposition internally and does not terminate in pole-discharge. Formal shorthand: $\text{Inv}(x)$. Status: Derived shorthand. Differentiated articulation. A realizable condition that is not flat:

$\text{Diff}(x) := \text{Real}(x) \wedge \neg \text{Flat}(x).$

Status: Derived. Lowest-resolution invariant-bearing differentiated degree. The invariant occurs in realizable non-flat articulation at the least admitted degree after origin:

$\text{Degmin}(x) := \text{Inv}(x) \wedge \text{Diff}(x).$

Status: Derived. Genuinely higher-resolution differentiated degree. A higher articulation of the same invariant that adds an internal determination rather than merely renaming the lower one. Status: Derived target notion. Pre-metric orthogonal re-articulation. Admission of a new axis/domain irreducible to the exhausted current axis/domain, without yet assuming metric orthogonality. Status: Derived target notion. Same-axis saturation. The current admitted axis/domain is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same axis/domain. Any further purported articulation there would either be mere renaming or reparameterization of what is already borne, or would require a new irreducible axis/domain and therefore not belong to the same axis/domain. Same-axis saturation is not discharge, not resolution of the invariant, not collapse, not termination, and not a claim that nothing further happens. It is only exhaustion of lawful same-axis expression. Status: Derived target notion. Algebra unlock. The first licensing of ordered-pair description and lawful binary combination once two irreducible axes are admitted. Status: Derived consequence.

2.3 Primitive root laws

The present draft adopts the following primitive root laws.

PB1. Pole non-identity

$$0 \neq 1$$

PB2. Pole sterility

$$\text{Ster}(0) \wedge \text{Ster}(1)$$

PB3. Pole opposition

$$\text{Opp}(0, 1)$$

PB4. Realizability implies non-vacuousness and pole exclusion

$$\forall x \in C,$$

$$\text{Real}(x) \Rightarrow \text{NV}(x) \wedge x \neq 0 \wedge x \neq 1.$$

2.4 One-origin and no-prior-apartness discipline

The manuscript adopts the following root discipline.

- There is initially no prior basis under which the two poles are primitive separate things.
- Therefore the poles may not be read as two externally separated primitive things.
- Any admissible primitive origin must be read as one primitive condition bearing the opposition internally. This is a root discipline of the rewrite. It is not permission to smuggle in an unearned mediating substance.

2.5 Derived shorthand for the invariant

When convenient, write $\text{Inv}(x)$ for the following schematic situation:

- x is an admissible primitive origin-condition,
- x does not coincide with either isolated pole,
- the non-identical sterile poles are opposed,
- that opposition is internal to the one origin-condition,
- and the condition does not terminate in pole-discharge. This shorthand is mnemonic. It does not replace proof burden.

3. Foundational Construction: Forced Origin Architecture

3.1 Branch-scope witness

Definition 3.1.1.

An origin candidate is any $x \in C$ such that $\text{Orig}(x)$ holds.

Scope Convention 3.1.2 (The branch under study is realized).

This manuscript classifies the primitive origin of a realized falsifiable branch. Let ω denote the primitive origin-condition of the branch under study. Then

$\text{Orig}(\omega) \wedge \text{Real}(\omega)$.

This is not a detachable extra premise about arbitrary worlds. It is the witness fixed by the subject matter of the manuscript itself.

3.2 Nullity exclusion

Theorem 3.2.1 (Nullity exclusion).

Absolute nullity is not a viable realized root for a falsifiable law-bearing branch. Status: Proved in this CF. Proof. A falsifiable law-bearing branch must support at least the distinction between admissible and inadmissible articulation and between realized and unrealized burden. Absolute nullity supports no realized content and no realized distinction. Therefore it cannot serve as the realized root of the present branch. $\square$

3.3 Undifferentiated totality exclusion

Theorem 3.3.1 (Undifferentiated totality exclusion).

Absolute undifferentiated totality is not a viable realized root for a falsifiable law-bearing branch. Status: Proved in this CF. Proof. A falsifiable law-bearing branch requires some realized noncoincidence by which one admissible condition can fail to coincide with another or with itself in a relevant way. Absolute undifferentiated totality contains no internal noncoincidence and therefore cannot sustain differentiated realized articulation. Hence it cannot serve as the realized root of the present branch. $\square$

3.4 Shared sterility does not collapse the poles

Proposition 3.4.1.

The poles 0 and 1 are not identical even though both are sterile for the present branch. Status: Proved in this CF. Proof. By PB1, $0 \neq 1$. By PB2, both are sterile. Distinctness and shared sterility therefore coexist. $\square$

3.5 Exhaustive terminal classification at the primitive layer

Definition 3.5.1 (Terminal primitive pole-candidate).

A terminal primitive pole-candidate is a primitive origin-candidate considered only at terminal, flat, sterile resolution: no internal noncoincidence is available by which differentiated articulation could be admitted, and no added basis is available by which the candidate could be decomposed or separated.

Theorem 3.5.2 (No third terminal primitive pole survives).

Let t be any terminal primitive pole-candidate. Then exactly one of the following holds: 1. $t = 0$, 2. $t = 1$. In particular, no third terminal primitive pole survives. Status: Proved in this CF. Proof. At the primitive layer, classification can use only the weakest distinctions already on the page: whether any realized content is present at all, and whether any internal noncoincidence is available.

- If no realized content is present, the candidate is complete realized absence and so equals 0.
- If realized content is present, the candidate is non-vacuous; but if it is also terminal, flat, and primitive, then no internal noncoincidence is available by which differentiated articulation could be admitted, and no richer content may be imported from outside. Therefore realized content is present yet internally undifferentiated, so the candidate collapses to 1. There is no third case. Any purported third primitive terminal pole either

lacks realized content, possesses realized content without internal noncoincidence, or already contains richer articulated burden and therefore is not a terminal primitive pole-candidate. Hence no third terminal primitive pole survives. $\square$

3.6 Forced admissible origin theorem

Theorem 3.6.1 (Forced admissible origin theorem).

For the realized branch under study, the primitive origin-condition ω is one primitive condition bearing the opposition of 0 and 1 internally without terminal settlement. Equivalently,

$\text{Inv}(\omega)$. Status: Proved in this CF. Proof. By Scope Convention 3.1.2, ω is the primitive origin-condition of a realized falsifiable branch. By Theorem 3.2.1, $\omega \neq 0$. By Theorem 3.3.1, $\omega \neq 1$. By Theorem 3.5.2, there is no third terminal primitive pole. Therefore the realized branch-origin cannot be any terminally settled primitive pole at all. By Section 2.4, the primitive layer contains no prior basis under which two primitive poles could begin as externally separate things. So the relevant primitive opposition cannot be realized as two already-distinct primitive objects. By

Proposition 3.4.1, the two poles are genuinely distinct even though both are sterile in isolation. Since the realized

branch-origin cannot terminate in either isolated pole, cannot be a third terminal primitive pole, and cannot externalize the poles into two already-separated primitive things, the only admissible primitive possibility left is that the distinction between the two terminal poles is borne internally by one primitive origin-condition without terminal settlement. That is exactly the manuscript meaning of $\text{Inv}(\omega)$. $\square$

Corollary 3.6.2 (No simpler primitive survivor remains).

Relative to the inferential discipline of this manuscript, no simpler primitive survivor remains beyond: 1. isolated nullity, 2. isolated undifferentiated totality, 3. one primitive origin-condition bearing their unresolved internal opposition. The first two are excluded for any realized falsifiable branch, so only the third remains admissible. Status: Proved in this CF.

3.7 Invariant-bearing precludes flatness

Theorem 3.7.1 (Invariant-bearing is incompatible with flatness).

For any $x \in C$,

$\text{Inv}(x) \Rightarrow \neg \text{Flat}(x)$. Status: Proved in this CF. Proof. Flatness means that no realized internal noncoincidence is available by which differentiated articulation could be sustained. But $\text{Inv}(x)$ states precisely that unresolved pole-opposition is internally borne by one admissible origincondition without terminal discharge. A condition bearing unresolved internal opposition is therefore not flat. $\square$

3.8 First non-flat articulation follows directly from the forced origin

Theorem 3.8.1 (Breakdown of perfect flatness at the realized origin).

For the realized branch under study,

$\neg \text{Flat}(\omega)$. Status: Proved in this CF. Proof. By Theorem 3.6.1, $\text{Inv}(\omega)$. Apply Theorem 3.7.1. $\square$

Corollary 3.8.2 (Differentiated articulation is present at the realized origin).

For the realized branch under study,

$\text{Diff}(\omega)$. Status: Proved in this CF.

Proof.

By Scope Convention 3.1.2, $\text{Real}(\omega)$. By Theorem 3.8.1, $\neg \text{Flat}(\omega)$. Apply the definition of Diff . $\square$

Corollary 3.8.3 (Lowest-resolution invariant-bearing differentiated degree is witnessed).

In witness form, the present branch satisfies

$\exists x \in C \text{ Orig}(x) \wedge \text{Degmin}(x)$,

with witness $x = \omega$. Status: Proved in this CF. Proof.

By Theorem 3.6.1 and Corollary 3.8.2, $\text{Inv}(\omega) \wedge \text{Diff}(\omega)$, hence $\text{Degmin}(\omega)$. $\square$

3.9 Section summary

At this stage the manuscript has:

- proved exclusion of isolated nullity,
- proved exclusion of isolated undifferentiated totality,
- proved distinctness of the poles despite shared sterility,
- proved exhaustive terminal classification at the primitive layer,
- proved that no third terminal primitive pole survives,
- proved that the only admissible primitive origin left is one origin-condition bearing unresolved pole-opposition,
- proved that no simpler primitive survivor remains for a realized falsifiable branch,
- proved incompatibility of invariant-bearing with flatness,
- proved differentiated articulation at the realized origin,
- and proved the lowest-resolution invariant-bearing differentiated degree.

4. Main Theorems and Proofs

4.1 Dependence-ranked articulation chain

The articulation chain must now be read in dependence-ranked form: 1. one admissible origin-condition carries unresolved internal pole-opposition, 2. a lowest-resolution invariant-bearing differentiated degree is admitted, 3. the question becomes whether that degree admits a genuinely higher-resolution differentiated degree under the same invariant, 4. if such a degree is admitted, the lower degree is no longer sufficient as sole determination, 5. within an admitted axis/domain, every genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that axis/domain, 6. once no genuinely new lawful same-axis articulation remains, same-axis saturation holds,

7. if same-axis saturation obtains while discharge remains impossible, orthogonal re-articulation is forced, 8. the first such forced orthogonal re-articulation closes the move into 2D, 9. only then is algebra licensed.

This is not chronology. It is dependency order.

4.2 HRDCL - character of a genuinely higher-resolution degree

Theorem 4.2.1 (Higher-Resolution Degree Character Lemma).

If a lowest-resolution invariant-bearing differentiated degree admits a genuinely higher-resolution differentiated degree under the same invariant, then:

1. the lower degree is no longer sufficient as the sole determination of the state, 2. an additional internal determination is present, 3. metric geometry, carrier structure, and differentiability still do not follow from this theorem alone.

Status: Proved in this CF. Proof. A genuinely higher-resolution differentiated degree is not a vacuous restatement of the lower one but a more articulated inhabitation of the same invariant. If the lower degree remained sufficient as sole determination, no added internal determination would have been introduced and the alleged higher degree would collapse into renaming. That contradicts genuineness. Hence the lower degree is no longer sufficient and an additional internal determination is present. But this alone does not state metric separation, carrier structure, or differentiability. $\square$

4.3 Invariant-bearing is constitutive of admitted articulation

Theorem 4.3.1 (Non-bearing is contradictory).

Let x be an admitted branch articulation. Then $\neg \text{Inv}(x)$ is contradictory. Status: Proved in this CF. Proof. Assume x is an admitted branch articulation and $\neg \text{Inv}(x)$. Then the primitive unresolved opposition is not borne in x . That is discharge. Terminal discharge resolves the branch-origin burden toward a terminal pole. By Theorem 3.5.2, no third terminal option survives; so discharge collapses toward either 0 or 1. But collapse toward 0 contradicts Theorem

3.2.1 and collapse toward 1 contradicts Theorem 3.3.1 for any realized falsifiable branch. Therefore non-bearing is contradictory. $\square$

Corollary 4.3.2.

Invariant-bearing is not an added persistence claim above admission. It is constitutive of admission. Status: Derived structural consequence.

4.4 HRDAP - admission of a genuinely higher-resolution degree

Theorem 4.4.1 (Higher-Resolution Degree Admission Principle).

At least one origin-bearing Degmin admits a genuinely higher-resolution differentiated degree under the undischarged primitive invariant. Status: Proved in this CF. Proof. Assume HRDAP fails. Then every origin-bearing Degmin is exhaustively complete as the terminal articulation of the invariant in that degree. Exhaustive completion is terminal settlement. Terminal settlement resolves the unresolved pole-opposition. By Theorem 3.5.2, no third terminal option survives. So terminal settlement collapses toward either 0 or 1, both excluded by Theorems 3.2.1 and 3.3.1 for any realized falsifiable branch. Contradiction. Therefore exhaustive completion of Degmin is false, and at least one origin-bearing Degmin admits a genuinely higher-resolution differentiated degree. $\square$

4.5 Contextual noncoincidence and apartness

Theorem 4.5.1 (Contextual Noncoincidence).

Two distinct higher-resolution determinations of the same invariant at the same articulation level are not identical as the same determination. Status: Proved in this CF. Proof. Suppose two allegedly distinct higher-resolution determinations $D1$ and $D2$ are identical. Then no new determination exists at that level. The level is exhaustively complete by a single determination. But by the contradiction used in

Theorem 4.4.1, exhaustive completion forces terminal settlement and collapse toward an excluded pole. Therefore $D1 \neq D2$. $\square$

Corollary 4.5.2 (Apartness is licensed).

Once contextual noncoincidence is proven between same-level determinations, primitive apartness of determination is licensed. Status: Proved in this CF. Proof. Apartness here means only that distinct same-level determinations are not the same determination. Theorem 4.5.1 gives exactly that. No metric structure is implied. $\square$

4.6 ORS - pre-metric orthogonal rotation structure

Theorem 4.6.1 (Orthogonal Rotation Structure, pre-metric form).

Let $D1$ and $D2$ be two contextually noncoincident higher-resolution determinations of the same invariant at the same articulation level. If the current axis/domain of articulation is exhaustively saturated by $D1$, then admissibility of $D2$ requires a new axis/domain irreducible to the saturated one. This irreducible re-articulation is the first pre-metric orthogonal rotation.

Status: Proved in this CF. Proof. By Theorem 4.5.1, $D1$ and $D2$ are not identical. Suppose nevertheless that $D2$ is only further expression along the same exhaustively saturated axis/domain as $D1$. Then no genuinely new admissible determination is available there; the axis/domain is already complete. That is same-domain exhaustion. Since non-bearing is contradictory by Theorem 4.3.1, the invariant cannot terminate there. Therefore the admissibility of $D2$ cannot be realized as more of the same exhausted axis/domain. A new axis/domain, irreducible to the first, is required. This is exactly the manuscript meaning of pre-metric orthogonal re-articulation. $\square$

4.7 Same-axis saturation and forced orthogonal re-articulation

Definition 4.7.1 (Same-axis saturation).

The current admitted axis/domain is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same axis/domain. Any further purported articulation there would either be mere renaming or reparameterization of what is already borne, or would require a new irreducible axis/domain and therefore not belong to the same axis/domain. Clarification. Same-axis saturation is not discharge, not resolution of the invariant,

not collapse, not termination, and not a claim that nothing further happens. It is only exhaustion of lawful same-axis expression.

Theorem 4.7.2 (Exhaustive bearing within an admitted axis/domain).

Within a currently admitted axis/domain, any genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that same axis/domain. It cannot be omitted and still count as lawful completion of that axis/domain. Status: Proved in this CF. Proof. Let the current axis/domain be admitted. Suppose a genuinely new lawful same-axis invariant-bearing articulation remains available there and yet is left outside the articulation burden of that same axis/domain. Then the axis/domain is being treated as lawfully complete while lawful same-axis invariant-bearing articulation still remains unborne there. That is not completion of the current axis/domain at all; it is incomplete same-axis bearing falsely treated as closure. Since Theorem 4.3.1 makes non-bearing contradictory for admitted articulation, lawful same-axis invariant-bearing articulation cannot be discarded as though it were irrelevant to the current axis/domain. Therefore any genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that same axis/domain and cannot be omitted while claiming lawful completion there. $\square$

Theorem 4.7.3 (Saturation theorem).

Within a given admitted orthogonality class, once lawful same-axis expression of the invariant is exhaustively saturated, further admitted articulation in that same class is impossible without contradiction. Since terminal discharge is excluded, orthogonal re-articulation is forced. Status: Proved in this CF.

Proof. By Definition 4.7.1, same-axis saturation means no genuinely new lawful invariant-bearing articulation remains available within the current axis/domain. Suppose nevertheless that a further admitted articulation exists in that same class. If it is genuinely new and lawful, then Definition 4.7.1 was false and the axis/domain was not saturated after all. If it is not genuinely new, then it is only renaming or reparameterization of what is already borne and therefore not a new admitted articulation in that class. Contradiction. Therefore no further admitted articulation exists there once same-axis saturation holds. But if admitted articulation ceased altogether, the invariant would have no borne expression and would be discharged, contradicting Theorem 4.3.1. Hence further articulation must occur, but not in the saturated class. By

Theorem 4.6.1, it is forced to occur as orthogonal re-articulation. $\square$

4.8 0D $\rightarrow$ 1D $\rightarrow$ 2D closure

Definition 4.8.1 (Stage labels).

- 0D: the primitive origin stage, prior to admitted axis-specific articulation.
- 1D: the first admitted axis/domain of non-flat invariant-bearing articulation.
- 2D: the first stage at which a second irreducible axis/domain has been admitted by orthogonal re-articulation.

Theorem 4.8.2 (0D $\rightarrow$ 1D).

The forced origin together with Theorem 3.8 closes the move from 0D to 1D. Status: Proved in this CF. Proof. At 0D, the invariant is borne at the primitive origin without yet resolved axis-specific articulation. By Theorem 3.8, the realized origin is non-flat and differentiated. That is the first admitted axis/domain of articulated expression. Hence 0D

$\rightarrow$ 1D is closed. $\square$

Theorem 4.8.3 (1D $\rightarrow$ 2D).

The same-axis saturation theorem together with ORS closes the move from 1D to 2D. Status: Proved in this CF. Proof. By Theorem 4.7.2, lawful completion of 1D requires exhaustive same-axis bearing of genuinely new lawful same-axis articulation. By Definition 4.7.1, that is exactly same-axis saturation. Same-axis saturation is not discharge by the clarification in Section 4.7, and discharge is excluded by Theorem 4.3.1. Therefore lawful closure of 1D cannot terminate there. By Theorems 4.6.1 and 4.7.3, further articulation must occur as irreducible orthogonal re-articulation into a new class. The first such new class is exactly 2D. Hence 1D $\rightarrow$ 2D is closed. $\square$

4.8A Axis-proper effective invariants at first and second admission

Definition 4.8.4 (Continuous same-axis bifurcation, pre-metric sense).

At the present stage, “continuous” means only that once a first axis/domain is admitted, lawful same-axis invariant-bearing articulation is indefinitely reiterable in that axis/domain until same-axis saturation is reached. Bifurcation

here is not an additional postulate. It is the direct expression of the primitive two-pole opposition along the admitted axis/domain: each iterated articulation step inherits the irreducible polar structure of the invariant itself, presenting two irreconcilable directions of continuation. No metric continuum, coordinate parameter, calculus, or topology is implied.

Theorem 4.8.5 (Axis-proper effective invariant at 1D admission).

Once 1D is admitted, the invariant is not merely borne on an axis/domain as a passive container condition. The admitted 1D axis/domain immediately acquires a new effective invariant proper to that axis/domain: continuous same-axis bifurcative articulation under the inherited primitive invariant. Status: Proved in this CF.

Proof. By Theorem 4.8.2, 1D is the first admitted axis/domain of articulated invariant-bearing expression. By Theorem 4.3.1, invariant-bearing is constitutive of admission, so the admitted axis/domain cannot be neutral with respect to the invariant it bears. By Theorem 4.10.1, any realizable invariant-bearing articulation that remains non-exhaustive relative to the undischarged primitive invariant requires a further articulated condition again under that same invariant. Applied to first admission, lawful continuation in 1D is therefore not merely occupancy of a line but reiterated same-axis articulation under the same invariant. Since 1D is the first admitted axis/domain, this reiterated same-axis articulation is proper to that axis/domain itself. By Definition 4.8.4, this is continuous same-axis bifurcation. It is therefore a new effective invariant proper to 1D, borne together with the inherited primitive invariant. $\square$

Corollary 4.8.6 (1D bears the invariant twice).

At 1D, the branch bears the invariant in two non-identical but inseparable senses: first as inherited primitive unresolved burden, and second as the axis-proper effective invariant of continuous same-axis bifurcation. Status: Proved in this CF.

Proof. Immediate from Theorem 4.8.5 together with the continuing force of Theorem 4.3.1. $\square$

Corollary 4.8.7 (Seriality and proto-number at 1D).

Because the admitted 1D axis/domain supports continuous same-axis bifurcative articulation until saturation, 1D is the first stage at which seriality, repeatable more/less, and proto-natural-number structure become licensed. No later number-theoretic completion is claimed here. Status: Derived structural consequence.

Proof. Continuous same-axis bifurcative articulation licenses reiterable lawful distinction along one admitted axis/domain. That is sufficient for serial order, repeatable succession, and more/less structure. Because the invariant cannot discharge, articulation along 1D is non-terminating; each bifurcative step generates a successor without ceiling, which is exactly proto-natural-number structure. $\square$

Theorem 4.8.8 (Inherited stack and second-axis effective invariant at 2D admission).

When 2D is admitted, the prior invariant stack is inherited rather than erased: the primitive invariant and the 1D effective invariant remain borne, and the newly admitted second axis/domain immediately acquires its own effective invariant proper to that axis/domain. Status: Proved in this CF.

Proof. By Theorem 4.8.3, 2D is admitted only by orthogonal re-articulation from a same-axis-saturated 1D that still cannot discharge the invariant. Therefore admission of 2D cannot annihilate what 1D had already lawfully borne; otherwise discharge or loss of borne invariant-content would occur, contradicting Theorem 4.3.1. The prior invariant burden is therefore inherited. But by the same logic used in Theorem 4.8.5, a newly admitted axis/domain is not a neutral container: once admitted, it immediately enters lawful invariant-bearing articulation proper to itself. Hence 2D inherits the prior stack and adds a new second-axis effective invariant. $\square$

Corollary 4.8.9 (Numerical relation-grid at 2D).

Because 2D inherits 1D serial bifurcation and adds a second axis/domain with its own axis-proper effective invariant, the first grid of numerical relation becomes possible at 2D. This is the structural reason algebra first becomes admissible only there. Status: Derived structural consequence.

Proof. Two admitted irreducible axes carrying inherited and new bifurcative burden jointly license pair-placement and cross-axis relation. A cross-axis relation is a determination whose specification requires reference to both axes simultaneously and cannot be decomposed into a determination on either axis alone; that is exactly what the irreducibility of the two axes from each other first makes possible at 2D. That is exactly the first grid of numerical relation presupposed by Theorem 4.9.1. $\square$

4.9 Algebra unlock

Theorem 4.9.1 (Algebra unlock at 2D).

Once 2D is admitted, ordered-pair description and lawful binary combination are licensed. This is the first algebraic unlock of the branch. Status: Proved in this CF.

Proof. By Theorem 4.8.3, two irreducible articulation axes/classes are admitted. With two irreducible axes, a determination may be specified by its two-axis placement rather than by a single-axis burden alone. This licenses ordered-pair description (a, b) relative to the admitted pair of axes. Once ordered pairs are licensed, lawful binary combination and comparison of coefficients on those axes are also licensed. That is the first algebraic regime of the branch. No richer algebraic structure is claimed here. $\square$

4.10 Recursive restatement of the invariant

Theorem 4.10.1 (Recursive Reapplication Principle).

If a realizable invariant-bearing articulation is non-exhaustive relative to the undischarged primitive invariant, then a further articulated condition is required that again stands under that invariant. Status: Proved in this CF. Proof. The contradiction used to establish HRDAP does not depend on the specific articulation level being Degmin . The only resources used were: realized branch admissibility, non-discharge of the invariant, terminal-pole exclusion, and the no-third-terminal result. Therefore any admitted articulation that still leaves the invariant non-exhaustive cannot be terminal. A further articulation is then required under the same invariant. $\square$

4.11 Section summary

At the end of Section 4, the theorem-bearing articulation burden is closed at the declared scope of CF000. The manuscript has established:

- higher-resolution degree character,
- constitutive invariant-bearing of admission,
- higher-resolution admission,
- contextual noncoincidence,
- apartness,
- pre-metric orthogonal rotation,
- same-axis saturation,
- the 0D -> 1D -> 2D route,
- the 1D axis-proper effective invariant,
- continuous same-axis bifurcation,
- 1D bearing of the invariant twice,
- seriality / proto-natural-number structure at 1D,
- inherited invariant stacking at 2D,
- the 2D axis-proper effective invariant,
- first numerical relation-grid structure at 2D,
- algebra unlock,
- recursive restatement.

No carrier, metric, QGT, or differentiability theorem has been introduced. Nothing in Section 4 licenses those later objects retroactively. Their absence here is not a gap; it is the stage boundary.

5. Derived Physical Consequences

5.1 A8 visibility note

Forward note for CF00. The same logic that forces orthogonal re-articulation when a domain saturates also makes the later A8 program visible in embryo. The present document does not prove the quantitative logarithmic law. What it proves is the structural source of it:

- one invariant,
- finite lawful expression within a currently admitted domain,
- saturation without discharge,
- orthogonal re-articulation into a new domain. This is why hierarchy becomes visible here without becoming primitive here.

5.2 Forward note on effective invariants

Derived structural consequence with forward note for CF00.

There is only one universal invariant at origin. New domains do not introduce independent primitive invariants. Rather, each admitted axis/domain inherits the origin invariant together with all previously realized effective invariants and adds a new effective invariant proper to that admitted axis/domain. Thus the invariant structure is cumulative and nested: primitive at origin, effective and inherited across successive articulations.

At the scope of the present closure, this rule is already theorem-bearing for the first two admissions. At 1D, the admitted axis/domain immediately acquires the effective invariant of continuous same-axis bifurcative articulation. At 2D, the prior stack is inherited and a new second-axis effective invariant is added. What remains deferred to CF00+ is not the existence of axis-proper effective invariants at 1D and 2D, but the later identification of richer mature forms carried by subsequent domains and later physical realizations.

5.3 Forward physical test

Forward note for later CFs. If later black-hole physics is read as a domain-relative saturation event, then the horizon/singularity complex should not represent invariant annihilation. It should instead represent exhaustion of lawful same-domain articulation together with forced orthogonal re-articulation into a new domain. Failure of invariant inheritance there would falsify that later interpretation.

5.4 Interpretive extension kept out of theorem burden

Terms such as “tachyonic instability,” “spinodal,” or later field-theoretic realization may be useful interpretively, but they are out of scope for the theorem-bearing spine of CF000.

6. Validation Logic and Evidentiary Support

6.1 Validation logic for theorem closure

The theorem-bearing closure of CF000 does not depend on the Aura runtime, curated territory tables, or any single simulation artifact. The theorem load is carried in Sections 3 and 4.

What would count against the formalism at this level includes:

- a simpler primitive survivor beyond the two terminal poles and one unresolved admissible origin,
- an admitted branch articulation that can cease to bear the invariant without contradiction,
- a same-domain saturation state that neither discharges nor forces orthogonal re-articulation,
- a successful derivation of 2D without a second irreducible axis.

6.2 Aura finite-domain consistency note

Empirical consistency note (non-load-bearing). The uploaded Aura territory bundle is consistent with the finite-domain reading of saturation and with the later A8 scaling program. From the curated territory timeline and regime summary:

- the run reaches a stable maximum of 9 active territories,
- the long saturation regime begins at $t = 3216$,
- the $N = 8$ regime lasts 913 ticks,
- the $N = 9$ regime lasts 14,037 ticks. For a carrier size $L = 5000$,

$\ln(5000) = 8.517$, so the observed depth ratio is $N_{\max} = 1.0567$. In $L \ln(5000)$ This is consistent with the claim that finite domains organize into hierarchies of logarithmic depth.

6.3 Late saturation and internal repartition

Empirical consistency note (non-load-bearing). The strongest late-run signature is not indefinite same-level growth but saturation followed by internal repartition. In the late territory/community tables:

- active territories remain at 9 in sampled late states,
- community count rises from 8 at $t = 17160$,
- to 9 at $t = 17220$,
- to 19 at $t = 17280$,
- then remains elevated. This supports the reading that once a finite domain reaches its top-level articulation budget, further invariant burden does not disappear and does not continue as naive same-level accumulation. It reappears as internal partition inside the saturated level.

6.4 Novelty concentration at maximal depth

Empirical consistency note (non-load-bearing). The curated regime summary shows increasing novelty with territory depth, reaching approximately 0.815 at $N = 9$. This is consistent with later A8-style expectations that information concentration intensifies at the deepest admitted interface structure.

7. Worked Example or Minimal Realization

7.1 Minimal realization setup

Use the weakest stage-correct realization compatible with the theorem spine:

- two terminal primitive poles 0 and 1,
- one realized origin witness ω with $\text{Orig}(\omega)$, $\text{Real}(\omega)$, and $\text{Inv}(\omega)$,
- one first admitted axis/domain A1 supporting same-domain differentiated determinations,
- a same-axis saturation condition $\text{Sat}(A1)$ meaning no genuinely new lawful invariant-bearing articulation remains available within A1, so further purported articulation there would be either mere renaming or would require a new irreducible axis/domain. No metric, carrier, graph, Laplacian, smooth support, or potential ansatz is assumed in this worked realization. It is a logical realization of the theorem spine only.

7.2 Expected behavior

The worked realization should exhibit: 1. exclusion of stable realized branch-collapse into either isolated pole, 2. non-flat articulation from invariant-bearing origin, 3. higher-resolution determination within the initial axis/domain, 4. same-axis saturation in the strict sense that no genuinely new lawful same-axis articulation remains available, without discharge, 5. need for an irreducible second axis/domain for further lawful articulation, 6. ordered-pair/algebraic description only after that second axis/domain is admitted.

7.3 Success and failure criteria

Success at the level of CF000 means:

- no third terminal primitive pole is required,
- 1D carries an axis-proper effective invariant rather than functioning as a neutral container,
- continuous same-axis bifurcation licenses seriality, repeatable more/less, and proto-natural-number structure at 1D,
- same-axis saturation counts only as exhaustion of lawful same-axis articulation and does not count as discharge,
- continued genuinely new lawful articulation in the exhausted axis/domain is impossible without contradiction,

- the second irreducible axis/domain is required for continued articulation,
- 2D inherits the prior invariant stack and immediately acquires its own axis-proper effective invariant,
- a first numerical relation-grid becomes possible only once that second irreducible axis/domain is admitted,
- algebraic ordered-pair description becomes meaningful only after the second irreducible axis/domain is admitted.

Failure would look like:

- a third terminal primitive survivor carrying realized branch burden,
- 1D functioning only as passive axis occupancy with no axis-proper effective invariant,
- no serial/proto-number consequence from continuous same-axis bifurcation at 1D,
- genuinely new lawful continued articulation in the same saturated axis/domain without contradiction,
- discharge without collapse,
- loss rather than inheritance of the prior invariant stack upon 2D admission,
- or algebraic regime appearing before the second irreducible axis/domain is admitted.

7.4 Algebraic witnesses of the invariant spine

The following three cases demonstrate that the invariant logic proved in Sections 3–4 is generative, not merely classificatory. Each case shows a familiar mathematical object arising directly from the CF000 spine without importing external mathematical machinery. These are not proofs of those objects; they are recognition claims showing the spine already contains their structural source.

7.4.1 The imaginary unit i as ORS expressed algebraically

Status: Derived structural consequence. Scope-correct for CF000.

The 2D algebra unlock (Theorem 4.9.1) licenses ordered pairs (a, b) relative to two irreducible axes. Within this structure, consider the question: what is the algebraic operator whose repeated application cycles through the two-axis structure?

By Theorem 4.8.8, the two admitted axes are irreducible from each other and carry the inherited invariant opposition between them. An operator R acting on ordered pairs that rotates the pair by one axis-step must satisfy:

- $R(1, 0) = (0, 1)$ (maps the first axis to the second),
- $R(0, 1) = (-1, 0)$ (maps the second axis back, inverted, because the two poles are irreconcilable and returning along the same path reaches the opposite pole),
- therefore $R^2(1, 0) = (-1, 0)$, i.e. $R^2 = -1$ on the first-axis unit.

This is exactly the defining property of the imaginary unit: $i^2 = -1$. The operator R is i . Its existence is not an external postulate; it is the first algebraic expression of ORS (Theorem 4.6.1) on the 2D ordered-pair structure. The reason $i^2 = -1$ rather than $+1$ is that returning along the orthogonal rotation path reaches the opposite pole of the invariant, not the same one; the two-pole irreconcilability of the primitive invariant is precisely what prevents $R^2 = +1$. Complex arithmetic is therefore the natural algebra of a two-axis invariant-bearing structure — not an assumption imported into it.

7.4.2 $0! = 1$ as the identity element of proto- $\mathbb{N}$

Status: Derived structural consequence. Scope-correct for CF000.

Corollary 4.8.7 licenses proto-natural-number structure at 1D: a non-terminating sequence of successor steps generated by the two-pole bifurcation of the invariant. Within this structure, the factorial $n!$ counts ordered arrangements of n successor-elements. For $n = 0$, there are no elements to arrange. The number of ways to arrange nothing is exactly one: the empty arrangement.

This is not a convention. The empty arrangement corresponds to the identity element of the binary combination licensed by Theorem 4.9.1. Specifically, the 2D algebra unlock licenses lawful binary combination; every such combination

structure requires an identity element under which combination returns the original. The identity element under ordered arrangement is the empty arrangement, which is unique. The count of unique empty arrangements is therefore 1.

Thus $0! = 1$ is forced by the identity-element requirement of the algebraic structure generated at 2D, not stipulated. The pole mark **1** (absolute undifferentiated totality, excluded as terminal but preserved as the identity of combination) is the structural ancestor of this result.

7.4.3 Non-scope note: the Pythagorean theorem

Status: Forward note. Out of scope for CF000.

The Pythagorean theorem $a^2 + b^2 = c^2$ requires a metric on the carrier domain, which in turn requires the carrier construction and the induced metric sector derived from the QGT. Both belong to CF00 and CF01 respectively. The theorem is therefore not within CF000's scope boundary and is not demonstrated here. It is listed only to mark the stage boundary explicitly: the metric structure it requires is downstream of the handoff carried in Section 9.5.

8. CFN Pairing and Executable Traceability

No paired CFN is required for theorem closure of CF000. If a future CFN is created, it must mirror this document rather than repair it.

8.1 Prescribed CFN traceability plan

CF section

Future CFN segment

Quantities instantiated

Diagnostics emitted

Sec. 3 root elimination

root elimination

terminal poles, witness ω

pole-exclusion checks

Sec. 4 articulation chain

articulation chain

Degmin , higher-resolution witnesses

admission/contradiction logs

Sec. 4 orthogonal closure

orthogonal rotation

two irreducible axes

saturation + irreducibility diagnostics

Sec. 6 empirical notes

aura consistency

territory depth, plateau, repartition

depth-ratio and novelty summaries

Important rule: a future CFN may numerically witness these claims, but it may not supply missing theorem burden.

9. Assumptions, Limits, and Open Boundaries

9.1 Assumptions used in the derivation

- the branch under study is realized,
- the primitive layer admits only the two terminal poles as terminal primitive extremes,
- no prior apartness or external two-thing reading is available at origin,
- terminal discharge into either isolated pole is excluded for the realized branch,
- later-stage machinery is forbidden from prepaying root burden.

9.2 Limits of applicability

This CF closes only the branch-level route $0D \rightarrow 1D \rightarrow 2D$. It is not a carrier construction, not QGT, not metriplectic field dynamics, and not a differentiability proof.

9.3 Claims established here

- forced admissible origin,
- incompatibility of invariant-bearing with flatness,
- differentiated articulation and Degmin,
- constitutive invariant-bearing of admitted articulation,
- HRDCL,
- HRDAP,
- contextual noncoincidence and apartness,
- pre-metric orthogonal rotation,
- same-axis saturation,
- $0D \rightarrow 1D \rightarrow 2D$ closure,
- the 1D axis-proper effective invariant,
- continuous same-axis bifurcation,
- 1D bearing of the invariant twice,
- seriality / proto-natural-number structure at 1D,
- inherited invariant stacking at 2D,
- the 2D axis-proper effective invariant,
- first numerical relation-grid structure at 2D,
- algebra unlock,
- recursive restatement.

9.4 Claims not established here

- carrier-family construction,
- QGT,
- named J, M, or $J \oplus M$ split,
- metric or Laplacian derivation,
- differentiability,
- exact A8 constants and interface fractions,
- the full later tower of effective invariants beyond the 1D and 2D instances closed here,
- tachyonic field realization.

9.5 Open boundaries and future work

- CF00 must begin from the inherited post-2D invariant stack proved here rather than from a primitively assumed carrier,
- CF00 must derive the first lawful host/carrier consequences of that closure,
- the A8 program should formalize the quantitative scaling law on later carriers,
- the fuller later tower of effective invariants beyond the 1D and 2D instances closed here belongs to CF00+,
- physical field-theoretic realization belongs later than CF000.

10. Integration with Broader VDM Theory

- CF000: 0D -> 1D -> 2D, including the first axis-proper effective invariants at 1D and 2D, serial/proto-number licensing at 1D, and numerical relation-grid/algebra unlock at 2D, - CF00: begins from the inherited post-2D invariant stack and derives the first lawful host/carrier consequences of that closure en route to QGT, - CF01: develops the QGT / 3D -> 4D route.

For the present manuscript, the correct integration discipline is:

- CF000 closes the origin invariant, the first articulation, higher articulation, the first axis-proper effective invariant at 1D, and the first forced orthogonal move into 2D, - that 2D closure inherits the prior stack, adds a new second-axis effective invariant, and is what licenses the first numerical relation-grid and algebra, - CF00 then begins from that completed stacked handoff and derives the next lawful host/carrier layer rather than taking carrier structure as primitive.

11. References and Provenance

Internal reference objects relevant to this CF include:

- H000 primitive bifurcation proposal materials,
- the doctrinal note on the single fundamental primitive bifurcation law,
- A8 proposal materials as later-regime quantitative program,
- CF00 as downstream carrier/QGT formalism,
- Aura runtime consistency tables as non-load-bearing evidence notes. These remain provenance and integration references only. They do not replace derivation already written here.

Appendix A. Symbol Table

Symbol

Meaning

Status

Absolute nullity

Primitive

Absolute undifferentiated totality

Primitive

C

Logical domain of candidate realization conditions

Primitive setup

Orig(x)

Origin-candidate predicate

Primitive

Ster(x)

Sterility predicate

Primitive

Flat(x)

Flatness predicate

Primitive

NV(x)

Non-vacuousness predicate

Primitive

Real(x)

Realizability predicate

Primitive

Opp(x, y)

Opposition predicate

Primitive

Inv(x)

Primitive invariant carriage shorthand

Derived shorthand

Diff(x)

Differentiated articulation

Derived

Degmin (x)

Lowest-resolution invariant-bearing differentiated degree

Derived

ω

Realized branch-origin witness

Witness

1D axis-proper effective invariant

Continuous same-axis bifurcative articulation

Derived

Continuous same-axis bifurcation

Indefinitely reiterable polar articulation along 1D

Derived

Serial / proto- $\mathbb{N}$ structure

Successor structure generated by non-terminating 1D bifurcation

Derived structural consequence

Inherited invariant stacking

Preservation of the prior invariant stack at each new axis/domain admission

Derived

Numerical relation-grid invariant

2D axis-proper effective invariant of inter-axis irreducible numerical relation

Derived

First numerical relation-grid

Cross-axis determination structure first available at 2D

Derived structural consequence

Appendix B. Dependency Audit

Primitive root package

- > nullity and undifferentiated-totality exclusion
- > no third terminal primitive pole
- > forced admissible origin
- > invariant-bearing incompatible with flatness
- > differentiated articulation / Degmin
- > constitutive invariant-bearing of admission
- > HRDCL
- > HRDAP
- > contextual noncoincidence / apartness
- > 1D admission
- > 1D axis-proper effective invariant
- > continuous same-axis bifurcation
- > 1D bearing of the invariant twice
- > serial / proto-natural-number structure
- > ORS
- > same-domain saturation
- > 2D admission by forced orthogonal re-articulation
- > inherited invariant stacking at 2D
- > 2D axis-proper effective invariant
- > first numerical relation-grid structure
- > algebra unlock
- > recursive restatement

No later-stage theorem in this document is intended to carry backward support for an earlier theorem. Empirical notes in Section 6 are consistency checks only and do not support any theorem-bearing dependency.